RAG vs NON-RAG RESPONSE COMPARISON REPORT

Day 1- Assignment 2 – Generative AI

---

1. Introduction

This assignment compares the performance of a Large Language Model (LLM) when responding to queries in two different modes:

- **Without Retrieval (Non-RAG)** – responses generated using only pre-trained knowledge
- **With Retrieval (RAG)** – responses generated using additional external or retrieved data

The objective is to evaluate how retrieval improves response quality in terms of:

- Accuracy
- Relevance
- Hallucination (incorrect or made-up information)

---

2. Methodology

The comparison is conducted using 10 different queries covering:

- General knowledge
- Technical concepts
- Customer support scenarios
- Time-sensitive or specific information

For each query:

- A response is generated using Non-RAG
- A response is generated using RAG (with additional context)
- Both responses are compared and evaluated

---

3. Comparison Table

No	Query	Non-RAG Response	RAG Response	Accuracy	Hallucination	Result
1	What is the company's latest revenue?	Gives general financial info, no exact data	Provides specific revenue with details	Low vs High	High vs Low	RAG
2	Explain onboarding process	Generic explanation	Includes structured company steps	Medium vs High	Medium vs Low	RAG
3	Side effects of a specific medicine	General answer	Detailed and specific data	Low vs High	High vs Low	RAG
4	Explain Pythagorean theorem	Correct explanation	Same correct explanation	High vs High	Low vs Low	Tie
5	What is product roadmap?	Assumes features incorrectly	Gives actual planned features	Low vs High	High vs Low	RAG
6	Who is the CEO of a company?	May be outdated	Updated and verified info	Medium vs High	Low vs Low	RAG
7	Capital of India	Correct answer	Same correct answer	High vs High	Low vs Low	Tie
8	Write email to a client	Generic email	Personalized email with details	Medium vs High	Low vs Low	RAG
9	What happened in last meeting?	Cannot answer	Gives specific meeting details	Low vs High	None vs Low	RAG
10	What is machine learning?	Accurate definition	Same definition with extra detail	High vs High	Low vs Low	Tie

---

4. Analysis

4.1 Performance Differences

The comparison shows that RAG significantly improves responses in cases where:

- Specific or internal data is required
- Up-to-date information is needed
- Context-based answers are important

Non-RAG performs well mainly in:

- General knowledge questions
- Basic definitions
- Static factual information

4.2 Accuracy Comparison

- Non-RAG responses are often **general and sometimes incomplete**
- RAG responses are **more precise and context-aware**

This indicates that retrieval helps the model generate **fact-based answers instead of assumptions**.

4.3 Hallucination Analysis

- Non-RAG has a higher chance of **hallucination**, especially when information is missing
- RAG reduces hallucination by using **real retrieved data**

This makes RAG more reliable for real-world applications.

4.4 Use Case Understanding

RAG is more useful for:

- Business data
- Customer support systems
- Medical or technical queries
- Real-time information

Non-RAG is sufficient for:

- Basic knowledge
- Educational concepts
- Simple explanations

---

5. Key Findings

- RAG produced better results in **7 out of 10 queries**
- Both methods performed equally in **3 general knowledge queries**

- Non-RAG did not outperform RAG in any case

---

6. Conclusion

This comparison demonstrates that Retrieval-Augmented Generation (RAG) enhances the quality of LLM responses by providing accurate, relevant, and context-based answers.

While non-RAG is effective for simple and general queries, RAG is essential for handling complex, domain-specific, and real-time information.

Therefore, RAG is recommended for applications where **accuracy and reliability are critical**, especially in customer support and enterprise systems.